

Context

Under the most recent Accreditation Review by the Australian Medical Council (AMC) the RANZCP has been provided a condition under Standard 5

23. Systematically review the breadth of assessment methods with a view to reducing the burden of assessment on trainees and their supervisors. This includes an evaluation to determine reasons for the high prevalence of breaks in training undertaken in order to complete summative assessments, so that there is improved alignment of assessment requirements and program duration. (Standards 5.1 and 5.2)

Purpose of Review

An evidenced based approach to the concept of Burden of Assessment in postgraduate medical education is sought. Specifically, a defined approach to measurement.

Approach

Experts in the field were consulted for relevant terms and potential lead authors. A Google Scholar search was performed for the terms burden of assessment and medical education. As well as for authors Mary Ott and Eusang Ahn with the above related terms. References of identified papers were scrutinized for additional relevant papers.

It should be emphasized that this was not a systematic review of the literature and a scoping review would be the next step.

Results

10 papers have been identified in the bibliography as per below. All papers were generated from Canadian authors leading resident (postgraduate) Competency Based Medical Education (CBME) training programs in Canada. A number of papers utilized a grounded theory (qualitative) approach to understand themes around the perceptions of assessments and their potential burden. The majority of the literature in this area is more focused around the trainee or learner perspective (versus the supervisor/assessor or program perspective).

Key Observations

Lockyer et al

Is an often-cited consensus paper outlining core principles for assessment in CBME.

Key points include:

- CBME relies on a program of assessment with multiple methods of assessment and multiple assessors within an educational system.
- Assessment for learning is a driving principle.
- Faculty development is required to create a “shared mental model” of required learner behaviour and expected levels of performance.
- Variance in assessor rating is not all attributable to error, some variance reflects the lens through which an assessor sees a learner.

- The assessment form is an aid. The instrument is the assessor who conducts the assessment, who needs training in their role.

Ott et al

The authors outline an approach to understanding resident assessment burden proposing Self Determination Theory as an explanatory model:

1. Threats to Autonomy
 - a. Missed opportunities to self-regulate, i.e. residents not feeling in control of their own learning.
 - b. Lack of situational control, i.e. adding further stress when residents felt accountable for achieving EPAs when opportunities did not present themselves.
 - c. Comparative assessment, i.e. keeping up with the rest of the cohort and worrying about evaluations that are felt to not be helpful to progression and learning.
2. Threats to Relatedness.
 - a. Lack of trust, i.e. feeling unfairly judged by the competence committee.
 - b. Lack of time and energy, i.e. placing a strain on supervisor relationships and feeling uncomfortable asking for another assessment.
 - c. Lack of connection, i.e. when support from faculty and supervisors to complete EPAs was absent.
3. Threats to Competence.
 - a. Lack of clarity, i.e. feedback around assessments being unclear about what the resident has achieved and what is missing and overly focused on which assessments or EPAs have been completed.
 - b. Unrealistic expectations, i.e. where assessments are impractical because the experience is not available.
 - c. Forms vs feedback, i.e. having too much focus on completing the boxes and getting the ratings right versus providing useful feedback.

The authors provide some suggestions for improvement according to these themes.

1. Supporting Autonomy – Gradually providing learners the responsibility for seeking out feedback to self-regulate their own learning and seeing as entrustment as marking the beginning, not the end of competence.
2. Supporting Relatedness - Coaching meets the relational need for learners to feel supported and can help learners set mastery goals.
3. Supporting Competence – Utilizing EPAs for teaching as well as assessment.

Day et al

Authors showed that residents supported CBME in theory but identified 5 key themes with its implementation:

1. Administrative burden leading to frustration and anxiety.
 - a. Assessments were felt to be a time burden, often competing with clinical responsibilities.

- b. The need to complete frequent assessments and “chase down” supervisors to complete forms was seen as disrupting workflow and clinical responsibilities.
 - c. Anxiety could develop around not being able to meet all of the competencies within the given time frame
- 2. Learning overshadowed by focus on assessment.
 - a. A focus on assessment could detract from learning.
 - b. Although individual assessments were recognized as low stakes and formative, the ultimate summative purpose of assessments led to performance anxiety.
 - c. Narrowing focus around achieving a specific competency or checklist of assessments came at the detriment of other learning experiences.
- 3. Feedback quality on assessments was variable.
 - a. More frequent assessments and judgements would be worthwhile if residents received consistent, specific and constructive feedback along with coaching.
 - b. Rating scales were seen to prioritize “checking boxes” over meaningful feedback.
- 4. Gaming the system was common.
 - a. Entrustment decisions and judgements varied considerably between supervisors.
 - b. Prior knowledge of supervisor preferences was shared beforehand amongst residents with the express purpose of being deemed sufficiently competent on assessments.
 - c. Other residents would purposefully seek out tasks they would succeed at or lenient supervisors.
 - d. Other residents described participating in a “social game” to ingratiate themselves with supervisors to improve outcomes.
- 5. Faculty engagement and support was valued.
 - a. Experiences were valued where supervisors were proactive in initiating assessment encounters and engaged in coaching.
 - b. Residents valued when faculty sought to understand and address barriers to obtaining program requirements or offered flexibility.
 - c. Longitudinal relationships with faculty and supervisors were valued.

Ahn et al

Authors looked at residents’ real world experience and perspectives of EPAs. Residents were quite polarized in either viewing EPAs as a valuable opportunity for professional growth or an onerous requirement that interfered with learning.

These perspectives were informed by the following 3 themes:

- 1. Program administration and its perceived messaging to residents. In particular issues around perceptions that the focus was on the number of EPAs achieved.
- 2. Faculty assessors and their perceived degree of engagement or “buy-in”. The “gold standard” for EPAs to be found useful was to have assessments done in real time with both resident and assessor present. Residents who did not find EPAs useful describe them being done not in person and several days after the encounter.
- 3. Learner’s perceptions and behaviour. Residents who felt more supported by their program reported being more in charge of their own education and training and saw EPAs as more valuable.

Discussion points included:

1. Learner engagement. Assessment requires engagement of both assessor and learner. They proposed that a focus on “EPAs by numbers” may be stretching and stressing residents unnecessarily by setting unrealistic goals. They propose Vygotsky’s zone of proximal development as a model for considering EPA progression and milestones.
2. EPA selection process. EPAs should be seen as a beneficial roadmap for developing competence. Selective triggering or “cherry-picking” of EPAs can be effective if it allows learners to quickly demonstrate proficiency in some EPAs, giving them greater time and confidence to focus on tasks that require improvement. This is opposed to a system where learners can “cherry-pick” EPA assessments with the sole purpose of passing onto the next stage of training.
3. Reliance on self-assessment. Learners should not be required to complete assessment forms entirely on their own, particularly feedback sections.
4. Reclaiming authenticity. Efforts should be made to weave EPAs assessments into the normal daily workflow and culture.

Martin et al

Authors highlight that EPAs and entrustment scales may support assessment for learning in postgraduate medical education, but their successful implementation raises a number of tensions and dilemmas that need careful management to support competence development.

1. Standardization versus flexibility. Residents often experience difficulties matching their actual learning experiences to a predefined set of EPAs. Residents report feeling frustrated if EPAs do not capture the breadth, nuance or complexity of day-to-day clinical tasks. As well as a worry about EPAs becoming “tick-box’ exercises.
2. Learning versus performing. Residents reported noticing an increase in the frequency of direct observations and feedback since introduction of EPAs. Residents highlighted a risk of “assessment fatigue” as well as risks of being strategic around EPA choice if assessment opportunities were required to “pass”. Residents also conveyed concern that once an EPA has been completed the tendency might be to move onto other EPAs when further competency is required around that particular EPA.
3. Words versus numbers. Residents did not report that a new entrustment scale was a significant change from prior numeric scales. Entrustment scales were not seen to be very meaningful to residents. Residents consistently emphasized that verbal feedback is more important than the scores.

L Donna et al

Authors determined that direct observation whilst widely endorsed as an educational strategy created significant anxiety in the learner. This could then lead to challenges in authenticity with the resident changing their normal approach to a “textbook approach”, generating uncertainty about their role and raising questions about whether observers saw an authentic portrayal of the residents knowledge and skills.

Szulewski et al

Authors looked at how 8 residency programs were adapting to assessor burden. They identified 3 themes:

1. Disparate mental models of assessment processes in CBME.
 - a. Common adaptations included revising entrustment scales for ease of use and clarity, revisiting progression stages and EPAs to address fragmentation, faculty development, sharing tips and tricks.

- b. Unique adaptations included revising resident resources, improving resident participation on program committees, promoting an end of rotation discussion between residents and faculty about how CBME is working, including reviewing the entrustability scales.
- 2. Challenges in workplace-based assessment processes.
 - a. Common adaptations were encouraging direct observation through dedicated clinical time for faculty, leveraging other assessment strategies beyond EPA forms (e.g. performance reviews), faculty development around coaching, reviewing software functionality and initiating assessments proactively according to EPA needs.
 - b. Unique adaptations included highlighting for faculty and residents assessment information that is most useful, raising awareness of assessment opportunities, offering incentives and competition for assessors.
- 3. Challenges in performance review and decision making.
 - a. Common adaptations included enhancing reporting functionality in the reporting software and offering professional development opportunities for faculty.
 - b. Unique adaptations included enhancing transparency of competence decisions with a nominated resident representative, hosting retreats to review competence and EPAs and develop a shared faculty understanding.

Mann et al

Authors were looking at residents' perceptions prior to entering a CBME program and identified four themes.

1. Rumours contribute to perceptions of CBME. Rumours came from a variety of sources but mainly peers and near peers, although none indicated they looked for official information on the college website.
2. Perceptions of assessment and feedback. It was anticipated that implementation of CBME would improve feedback and asking for feedback would become more common place. Time to complete assessments was identified as a significant concern. Clarity around competence was also a concern.
3. Perceptions of teaching and learning. It was felt that CBME should lead to increased clarity of learning objectives and ability to be more self-directed in learning.
4. Implementation of CBME. Participants tended to express pessimism or uncertainty around implementation with logistical challenges most often cited.

Cheung et al

In a quantitative study in a quantitative study demonstrated that the average time taken to complete and Entrustable Professional Activity (EPA) assessment form was 3 minutes and 6 seconds, demonstrating faster completion times with increasing familiarity and repetition. Importantly a weakness of the study was that *"time added which many not be captured in this data includes time taken to consider resident performance prior to opening the assessment form as well as time taken discussing performance with the resident in person and/or coaching through weaknesses identified through the assessment process."*

Implications for RANZCP

The findings above are consistent with RANZCP developing a more mature approach to Competence, Entrustment, EPAs and workplace-based assessment.

The findings from the above papers:

- Highlight that the current model of trainees targeting 1 to 2 EPAs for completion in a 6-month rotation represents a disconnected approach to competency progression. The suggested shift in focus to requiring a set number of entrustability assessments and discussions is likely to be seen as more useful by trainees.
- Highlight the value of providing useful feedback and tying feedback into ongoing supervision and coaching as being a much more valuable area to focus on rather than collecting ratings on performance.
- Indicate that some trainees will view WBAs and EPAs as “ultimately summative” and be concerned and anxious about their performance and may engage in attempts to “game” the process.
- Indicate a risk in placing too much responsibility on trainees to initiate workplace-based assessments and suggest that our supervisors need to be trained and supported to take on a proactive approach, as well as be provided skills in providing useful feedback and ongoing performance coaching.

Conclusions

The burden of assessment in postgraduate medical education is a complex construct which most likely constitutes both a quantifiable aspect (e.g. amounts of time and resource inputs as well as opportunity costs), as well as a qualitative aspect, with trainee and assessor perceptions being important. Perceptions are also affected by both internal factors that relate to the training program and as well as external factors related to the clinical setting (learning environment) in which training sits.

Ott’s conceptualization of Burden of Assessment through the lens of Self-Determination Theory provides a useful framework for conceptualizing the concept of burden of assessment in order to permit regular collection of data from both trainees, supervisors and directors of training to attempt to measure the burden of assessment.

Suggested Questions

Rate Each Assessment Component

Use a ranking scale of 0-9

0 = totally disagree, 9 = totally agree

As trainees, supervisors, directors of training to rank each assessment from 0-9. The burden of assessment for this assessment...

Rate the Overall Assessment Program

Thinking about your overall experience of the RANZCP Program in terms of all the assessments you have had experience with so far how much would you agree or disagree with the following statements?

(totally disagree to totally agree)

In the RANZCP program:

[Autonomy Questions]

- the burden of assessments is fair
- the required assessments do not have a negative impact on my other roles
- trainees are in control of their own learning
- trainees are fairly accountable for their part in achieving assessment requirements
- supervisors are fairly accountable for their part in achieving assessment requirements
- training programs are fairly accountable for their part in achieving assessment requirements
- the RANZCP is fairly accountable for its part in achieving assessment requirements

[Relatedness Questions]

- the assessments are achievable
- there are sufficient opportunities to complete assessments
- the assessments provide a fair appraisal of a trainees knowledge, skills and performance
- there is enough time to complete assessments
- there is enough support available to complete assessments

[Competence Questions]

- the assessments provide useful feedback and allow trainees to improve
- the assessments are realistic and achievable
- the assessments assess the knowledge, skills and capabilities that trainees are required to demonstrate in their regular work
- there is an appropriate balance between assessment and feedback

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